# Problem A. Prime Date

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

We will call a date *prime* if each digit in its notation is either equal to 0 or is a prime number. In other words, only the digits 0, 2, 3, 5, and 7 may appear in the notation of a prime date.

Dates are written in the format `DD.MM.YYYY`. The day and month are always written with two digits, with a leading zero if necessary, and the year — with four digits.

Today is 18 October 2026.

Find the nearest prime date that will occur strictly after today.

## Input

This problem has no input data.

## Output

Print the nearest prime date after 18.10.2026 in the format `DD.MM.YYYY`.

## Example

| standard input | standard output |
|---|---|
| | 18.10.2026 |

## Note

The answer in the example is provided only to demonstrate the output format and is not correct.

# Problem B. Three Numbers

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

You are given three integers $a$, $b$, and $c$.

In one operation, you can choose exactly two of the three numbers and increase both of them by 1.

Find the minimum number of operations required to make all three numbers equal.

## Input

The only line contains three integers $a$, $b$, and $c$ ($1 \le a, b, c \le 10^9$).

## Output

Print one integer — the minimum number of operations.

## Example

| standard input | standard output |
|---|---|
| 1 2 3 | 3 |

## Note

For the numbers 1, 2, and 3, one possible sequence of operations is:

- increase the first and second numbers: $(1, 2, 3) \to (2, 3, 3)$;
- increase the first and third numbers: $(2, 3, 3) \to (3, 3, 4)$;
- increase the first and second numbers: $(3, 3, 4) \to (4, 4, 4)$.

Thus, 3 operations are sufficient.

# Problem C. Chocolate Bar Division

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

Miras and Alisher bought a rectangular chocolate bar of size $n \times m$. The chocolate bar consists of $nm$ identical squares.

Initially, the chocolate bar is one rectangular piece. In one break, you can choose any existing rectangular piece and break it along one of the lines between adjacent rows or columns of squares. As a result, the chosen piece is split into two rectangular pieces.

After several breaks, possibly zero, each resulting piece is given entirely either to Miras or to Alisher.

Miras and Alisher want to divide the chocolate bar as fairly as possible. That is, the difference between the number of squares received by Miras and Alisher should be as small as possible.

Find the minimum number of breaks required for such a division.

## Input

The only line contains two integers $n$ and $m$ ($1 \le n, m \le 10^6$) — the number of rows and columns of squares in the chocolate bar.

## Output

Print one integer — the minimum number of breaks after which the chocolate bar can be divided between Miras and Alisher as fairly as possible.

## Examples

| standard input | standard output |
|---|---|
| 3 3 | 2 |
| 4 6 | 1 |

# Problem D. Long Path

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1.5 seconds |
| Memory limit: | 256 megabytes |

You are given a directed graph with $n$ vertices. Exactly 3 edges leave each vertex, and exactly 3 edges enter each vertex. From any vertex, it is possible to reach any other vertex by moving along the directions of the edges.

Each edge has a positive integer weight. A route is a sequence of directed edges such that the end of each edge coincides with the start of the next one. Vertices and edges in a route may repeat. The weight of a route is the sum of the weights of all edges in it.

You are given $q$ queries. In each query, you are given an integer $k$. For each query, determine whether there exists a route with weight exactly $k$.

## Input

The first line contains two integers $n$ and $q$ ($2 \le n \le 100$, $1 \le q \le 2 \cdot 10^5$) — the number of vertices in the graph and the number of queries, respectively.

The next $3n$ lines contain three integers $u$, $v$, and $w$ ($1 \le u, v \le n$, $1 \le w \le 100$) — a directed edge from vertex $u$ to vertex $v$ with weight $w$. It is guaranteed that from any vertex it is possible to reach any other vertex, each vertex has exactly 3 outgoing edges, and each vertex has exactly 3 incoming edges.

The next $q$ lines contain one integer $k$ ($1 \le k \le 10^{18}$).

## Output

For each query, print `YES` if there exists a route with weight exactly $k$, and `NO` otherwise. The case of letters does not matter.

## Example

| standard input | standard output |
|---|---|
| 5 6<br>1 2 7<br>1 3 5<br>1 4 5<br>2 3 5<br>2 4 5<br>2 5 5<br>3 4 5<br>3 5 5<br>3 1 5<br>4 5 5<br>4 1 5<br>4 2 5<br>5 1 5<br>5 2 5<br>5 3 5<br>5<br>7<br>10<br>12<br>13<br>17 | YES<br>YES<br>YES<br>YES<br>NO<br>YES |

# Problem E. Popular Topics

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1.5 seconds |
| Memory limit: | 256 megabytes |

Alan runs a blog and plans to publish exactly $k$ new videos.

He has $n$ different topics. Initially, the popularity of the $i$-th topic is $a_i$.

If Alan publishes a video on a topic with current popularity $x$, this video gets $x$ views. After that, the audience becomes less interested in this topic, and its popularity becomes $\lceil \frac{x}{2} \rceil$.

Alan can publish several videos on the same topic.

Help him determine the maximum total number of views that his next $k$ videos can get.

## Input

The first line contains two integers $n$ and $k$ ($1 \le n \le 2 \cdot 10^5$, $1 \le k \le 10^9$) — the number of topics and the number of videos.

The second line contains $n$ integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the initial popularity of the topics.

## Output

Print one integer — the maximum total number of views that $k$ videos can get.

## Example

| standard input | standard output |
|---|---|
| 4 5<br>10 7 4 1 | 30 |

## Note

One optimal way to choose topics is:

- choose the topic with popularity 10, after which its popularity becomes 5;
- choose the topic with popularity 7, after which its popularity becomes 4;
- choose the topic with popularity 5, after which its popularity becomes 3;
- choose one of the topics with popularity 4, after which its popularity becomes 2;
- choose the remaining topic with popularity 4, after which its popularity becomes 2.

The total number of views is

$10 + 7 + 5 + 4 + 4 = 30$.

It can be proved that it is impossible to get more than 30 views.

# Problem F. Three Cities

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 4 seconds |
| Memory limit: | 512 megabytes |

A film crew is shooting a movie in three cities: Astana, Almaty, and Tashkent.

There are $n$ days available for filming. If the crew shoots a scene on the $i$-th day, it takes $a_i$ minutes in Astana, $b_i$ minutes in Almaty, and $c_i$ minutes in Tashkent.

The crew has to shoot exactly $k$ scenes in each of the three cities. At most one scene can be shot on each day, and some days may be skipped.

The crew starts filming in Astana, then moves to Almaty, and finally to Tashkent. It cannot return to a previous city. Therefore, every day chosen for filming in Astana must be earlier than every day chosen for filming in Almaty, and every day chosen for filming in Almaty must be earlier than every day chosen for filming in Tashkent.

Find the minimum total amount of time required to shoot all $3k$ scenes.

## Input

The first line contains two integers $n$ and $k$ ($1 \le k \le 10^5$, $3k \le n \le 3 \cdot 10^5$) — the number of available days and the number of scenes that must be shot in each city.

The second line contains $n$ integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the time required to shoot a scene in Astana on each day.

The third line contains $n$ integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le 10^9$) — the time required to shoot a scene in Almaty on each day.

The fourth line contains $n$ integers $c_1, c_2, \ldots, c_n$ ($1 \le c_i \le 10^9$) — the time required to shoot a scene in Tashkent on each day.

## Output

Print one integer — the minimum total amount of time required to shoot all the scenes.

## Example

| standard input | standard output |
|---|---|
| 8 2<br>7 2 5 4 8 6 3 9<br>6 4 7 2 5 3 8 1<br>9 6 4 7 3 5 2 8 | 21 |

# Problem G. Stickers

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

Arman really likes stickers. Sometimes he even sticks them on top of each other.

At the ICPC Qualification, Arman received many stickers of two types: `kaspi` and `cpfed`.

After the competition, Arman took an empty strip and started sticking stickers onto it. He glued each sticker entirely onto some contiguous segment of the strip. The stickers could overlap each other. If a new sticker was glued on top of an old one, then the letters of the new sticker became visible.

As a result, Arman obtained the string $s$.

Determine whether Arman could have obtained the string $s$ using only the stickers `kaspi` and `cpfed`.

## Input

The only line contains the string $s$ ($1 \le |s| \le 2 \cdot 10^5$), consisting of lowercase Latin letters.

## Output

Print `YES` if Arman could have obtained the string $s$, and `NO` otherwise. The case of letters does not matter.

## Examples

| standard input | standard output |
|---|---|
| kaspifed | YES |
| kafed | NO |

# Problem H. Parity Sort

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

You are given an array $a$ of length $n$.

In one operation, you can choose an index $i$ ($1 \le i < n$) such that $a_i$ and $a_{i+1}$ have different parity, and swap them.

Determine whether it is possible to sort the array in non-decreasing order using some number of operations.

## Input

The first line contains one integer $n$ ($1 \le n \le 1\,000$) — the length of the array.

The second line contains $n$ integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the elements of the array.

## Output

Print `YES` if it is possible to sort the array in non-decreasing order using the described operations, and `NO` otherwise. The case of letters does not matter.

## Examples

| standard input | standard output |
|---|---|
| 5<br>2 1 4 3 5 | YES |
| 6<br>1 4 3 2 5 6 | NO |

# Problem I. Magicka

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

Andrey is playing a computer game called *Magicka*. In this game, you play as a sorceress named Loka, who creates spells by controlling magicks. On one of the levels, Loka needs to destroy a horde of orcs and goblins. Andrey wants to try Loka's recently learned spell, *Summon Pacman*, which summons a giant Pacman that eats everything in its path.

To cast the spell, Andrey has to arrange the magicks in the correct order.

A spell can be represented as a string consisting of the characters "$QWERASDFIV$". There are 8 basic magicks in the game: Water ($Q$), Life ($W$), Shield ($E$), Cold ($R$), Lightning ($A$), Death ($S$), Earth ($D$), and Fire ($F$). There are also 2 combined magicks: Steam ($V$) and Ice ($I$).

Andrey can add new basic magicks to the end of the string. While creating a spell, the following rules apply:

- Some magicks are incompatible with each other. The incompatible pairs are $Q - A$, $D - A$, $W - S$, $R - F$, and $E - E$. If two incompatible magicks are present in the string, both of them are removed;
- Some pairs of magicks create a new magick: $Q + R = I$, $I + F = Q$, $F + Q = V$, $V + R = Q$. If two magicks interact according to one of these rules, both of them disappear, and a new magick appears at the position of the left one.

When a new magick is added, the string is scanned from right to left until the first magick that can interact with it is found. If there is no such magick, the new magick is appended to the end of the string.

For example, if Andrey adds the magick "$E$" to the string "$QESD$", the resulting string will be "$QSD$".

If a magick "$Q$" is created inside the string as a result of an interaction, the search for the next interaction is performed in a special way: first, the magicks to the right of the newly created "$Q$" are checked from left to right. If no suitable magick is found there, the magicks to the left of it are checked from right to left.

For example, if Andrey adds the magick "$F$" to the string "$ASIA$", the resulting string will be "$AS$".

Andrey has already pressed several buttons, so he currently has some sequence of magicks selected. Help Andrey add basic magicks so that he obtains the spell *Summon Pacman*.

## Input

The first line contains a string $s$ ($1 \le |s| \le 10^5$) — the magicks that Andrey has already selected.

The second line contains a string $t$ ($1 \le |t| \le 10^5$) — the magicks required for the spell.

It is guaranteed that both $s$ and $t$ are valid possible states of the selected magicks.

## Output

Print a string consisting only of the characters "$QWERASDF$" — a sequence of basic magicks that Andrey should add in order to obtain the required spell.

You do not need to minimize the number of added magicks. You may print any valid sequence, but its length must not exceed $10^6$ magicks.

## Examples

| standard input | standard output |
|---|---|
| QVEQ<br>QQEW | ARW |
| WARWRAWIWA<br>WWWIQQ | FDDFFSRQQQ |

## Note

$QVEQ \to QVE \to QQE \to QQEW$

$WARWRAWIWA \to WARWRAWW \to WARWRWW \to WRWRWW \to WRWWW \to WWWW \to WWW \to WWWR \to WWWI \to WWWIQ \to WWWIQQ$

# Problem J. Two Trees

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

You are given two trees on the same $n$ vertices, numbered from 1 to $n$.

A segment $[l, r]$ is called good if the vertices $l, l+1, \ldots, r$ form a connected subgraph in each of the two trees.

You need to split the sequence of vertices $1, 2, \ldots, n$ into several good segments. Each segment must contain at least two vertices.

Find the maximum possible number of segments in such a partition. If it is impossible to split the sequence in the required way, output $-1$.

## Input

The first line contains an integer $n$ ($2 \le n \le 3 \cdot 10^5$) — the number of vertices in each tree.

The next $n-1$ lines describe the edges of the first tree. Each line contains two integers $u$ and $v$ ($1 \le u, v \le n$, $u \ne v$), meaning that vertices $u$ and $v$ are connected by an edge.

The next $n-1$ lines describe the edges of the second tree in the same way.

It is guaranteed that both given graphs are trees.

## Output

Output a single integer — the maximum number of good segments into which the sequence $1, 2, \ldots, n$ can be split.

If such a partition does not exist, output $-1$.

## Example

| standard input | standard output |
|---|---|
| 8<br>1 3<br>3 2<br>2 4<br>2 5<br>5 6<br>5 7<br>7 8<br>1 4<br>1 2<br>2 5<br>5 6<br>2 3<br>3 7<br>7 8 | 3 |

## Note

In the sample, the sequence can be split into three good segments: $[1, 4]$, $[5, 6]$, and $[7, 8]$. In all three cases, the corresponding vertices form a connected subgraph in each of the two trees. It is impossible to split the sequence into four segments.

# Problem K. ICPC Rounds

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 256 megabytes |

This year, the ICPC rounds in Kazakhstan take place on unusual dates:

- the 1/8 final — on October 18;
- the 1/4 final — on November 14;
- the 1/2 final — on December 12.

An alien learned about this and decided to hold a competition on his planet in a similar way.

On his planet, a year consists of $n$ months. Each month consists of $d$ days, and each week consists of $w$ days. The last $k$ days of each week are weekends. The first day of the year is the first day of the week.

The alien wants to hold all three rounds of the competition this year. To do this, he needs to choose three months $x$, $y$, and $z$ such that $x < y < z$, and hold:

- the 1/8 final — on the 18-th day of month $x$;
- the 1/4 final — on the 14-th day of month $y$;
- the 1/2 final — on the 12-th day of month $z$.

Each of the three rounds must take place on a weekend.

For each test case, find any suitable months $x$, $y$, and $z$, or determine that it is impossible.

## Input

The first line contains one integer $t$ ($1 \le t \le 1000$) — the number of test cases.

Each test case consists of one line containing four integers $n$, $d$, $w$, and $k$ ($1 \le n \le 10^9$, $18 \le d \le 10^9$, $1 \le k \le w \le 10^9$) — the number of months in a year, the number of days in a month, the number of days in a week, and the number of weekend days at the end of each week, respectively.

## Output

For each test case, output three integers $x$, $y$, and $z$ ($1 \le x < y < z \le n$) if it is possible to hold all three rounds on weekends.

The 18-th day of month $x$, the 14-th day of month $y$, and the 12-th day of month $z$ must all be weekend days.

If no suitable months exist, output $-1$.

If there are multiple answers, you may output any of them.

## Example

| standard input | standard output |
|---|---|
| 5<br>4 18 4 1<br>5 18 4 1<br>3 18 7 7<br>20 31 6 2<br>12 29 5 1 | -1<br>2 4 5<br>1 2 3<br>1 4 6<br>4 5 8 |